

Mary Knapp^a and Kyle Hughes^b

^aMassachusetts Institute of Technology, Haystack Observatory, Westford, MA, United States, ^bJet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States

1 Introduction

First contact: the moment the whole team has been waiting for and working toward through the long years of development, integration, and testing. The beginning of in-space operations is perhaps the most exciting and nerve-wracking moment of a space mission. Has deployment gone as planned? Will all of the subsystems function properly? Will the payload return the expected science or technical data?

Despite the nerves and sleepless nights, in-space operations can be the most rewarding part of the CubeSat development process. Careful preparation, planning, and training are needed to maximize the likelihood of smooth operations. While it is impossible to prepare for every eventuality or prevent every failure or mistake, this chapter aims at providing useful pointers for CubeSat operators, both new and experienced.

1.1 Scope

This chapter will step through preparation for operations, first contact, commissioning, and nominal operations. The future landscape for CubeSat operations and incorporate lessons learned from past missions will also be discussed.

1.2 Operational models

Operational models for CubeSats are as many and varied as CubeSats themselves. For the purposes of this chapter, the focus will be on two models: integrated operations, where spacecraft bus and science/payload operations are handled by the same (or overlapping) team in the same physical location, and split operations, where spacecraft operations (everything but the payload) are separated from science and/or payload operations. In the “split” operation case, the bus and payload teams may or may not be colocated and have less personnel overlap than in the “integrated” case. This chapter is written primarily with the “integrated” case in mind but is intended to be general enough to be applicable to both.

If the operation teams are separate or “split,” special care should be taken to develop open lines of communication between the bus operator(s) and the payload/science operations team. CubeSats by their nature are tightly coupled systems that

do not easily lend themselves to the separated bus-instrument model used for larger spacecraft. The instrument or payload will be strongly affected by what the bus is doing and vice versa. For example, reaction wheel speeds above a certain value may generate electromagnetic interference (EMI) that affects the performance of the payload.

The science/payload team should cultivate an understanding of how the bus works and its idiosyncrasies. The bus operator should likewise have a strong grasp of the science or technical mission and function of the payload so that precious time in orbit can be used efficiently. Cross participation in meetings and reviews between the bus and science/payload teams will open these vital lines of communication early in the mission lifecycle, leading to smoother in-orbit operations. Preflight operational readiness tests (Section 2.2) are especially important for split teams so that they can practice working together effectively.

2 Preparation for operations

The success of in-space operations depends on the prelaunch (or predeployment) preparation of the team.

2.1 Preflight testing and design for operations

Preparation for operations begins during integration and test (I&T). Planning I&T with operations in mind will help avoid last-minute scrambling to develop or refine operations tools and procedures. Personnel overlap from the I&T team to the operations team is desirable, since the I&T team learns the spacecraft's quirks during the testing process. At minimum, I&T personnel should be "on call" during early operations. Elements for consideration for preflight testing and operations design are provided in the following sections.

The spacecraft communications subsystem is, of course, critical to operations as it is the only pathway to command the spacecraft in-orbit and have visibility into spacecraft state and health. As such, radio failures are a common cause of CubeSat "infant mortality" or early loss of mission. Thorough testing of all communication systems is therefore essential to successful in-orbit operations.

2.1.1 Recommended testing

Ground station compatibility tests with flight hardware in the loop are *critical* to successful first contact and operations. The flight radio (or radios) should be tested on site with the ground station(s) that the mission will use in flight. Ideally a full FlatSat^a with integrated, flight-like radio(s) should be part of compatibility testing. At minimum the flight radio, flight computer (C&DH system), and power subsystem should be in the loop for compatibility testing. On the ground system side, the same set of software tools and user

^a A FlatSat is a system intended for ground testing of the spacecraft; it has all subsystems, but may not have the same physical/mechanical layout as the spacecraft flight model. In other words, it is all of the spacecraft subsystems and software laid out on a lab bench for the purpose of hardware and software testing.

interfaces that will be used for in-space operations should be used during compatibility testing. Ground hardware, including amplifiers, modems, and computer terminals, should also be the same as what will be used at the ground station(s) for flight operations.

2.1.2 Ground station redundancy

A single ground station represents a single point of failure. If at all possible, multiple ground stations at geographically distributed locations should be included in the spacecraft communications license. Even if the baseline plan is to use just one ground station, others should be licensed for backup in case of an outage. Seasonal weather outages due to high winds, snow, thunderstorms, etc. are common and should be accounted for in operations planning. If the spacecraft will need frequent contact to maintain health, backup ground stations are essential. A backup ground station can also be used to determine whether communications anomalies are due to a problem with the primary ground station or with the spacecraft (see [Section 4.1.1](#)).

2.1.3 Real-time telemetry design

Quickly and accurately assessing the state and health of the spacecraft is essential for timely anomaly response during operations. Real-time telemetry is therefore a critical tool for operations. Real-time telemetry is a set of telemetry points that are streamed to the ground automatically during a ground station contact. A well-chosen set of real-time telemetry points will provide operators with a snapshot of subsystem and overall spacecraft health. Real-time telemetry is often a subset of all spacecraft housekeeping telemetry generated and recorded onboard; the most critical health indicators are selected for inclusion in the real-time stream.

Suggestions for what to include in real-time telemetry:

- spacecraft mode or state (safe, nominal, etc.);
- flight computer boot count and/or uptime;
- temperatures of any components with thermal sensitivity (e.g., batteries and payload);
- solar panel currents/voltages;
- attitude state, including reaction wheel speeds if applicable, voltages/currents, and quaternion from attitude determination;
- radio mode or state, boot count (if applicable), temperature, and currents/voltages;
- fault protection state, including last fault responded to;
- payload state, currents, voltages, and temperatures.

Real-time telemetry channels should be chosen during development and exercised during functional and environmental testing so that the I&T team and operations team become familiar with the telemetry indicators of spacecraft anomalies.

2.2 Operational readiness tests

Operational readiness tests (ORTs) are training scenarios designed to mimic real in-space operations. ORTs are opportunities for the operations team to practice both nominal commanding and anomaly response in a lower-stakes environment than real flight operations. The timing of ORTs will depend on the specific schedule needs of a

mission, but they often take place near the end of the integration and test (I&T) period or soon after spacecraft delivery and before launch or deployment (see [Section 2.2.3](#)).

ORTs last a few days to a week and cover both nominal and off-nominal operational scenarios. During a typical ORT day, the operations team “operates” the spacecraft during several simulated ground contact opportunities.

2.2.1 Setup and design of ORTs

ORTs should strive for the highest level of fidelity to real on-orbit conditions possible given the availability of flight or flight-like hardware and software. Having the real flight unit (if predelivery) or a high fidelity engineering unit or FlatSat (if post-delivery) in the loop is strongly preferred. The ground support software and hardware should also be in the loop. Telemetry playback from preflight testing is also an option if actual flight or flight-like hardware is not available. The operations team should follow the same procedures and use the same tools that will be used for in-orbit operations. ORTs should cover several days of operations so that the operations team can practice the daily decision-making cycle for spacecraft and payload activities and the cadence of contact opportunities.

2.2.2 Operational scenarios

A key goal of ORTs is to “stress test” the operations team and train them to make careful but timely decisions about how to respond to anomalous spacecraft behavior. The project manager, I&T manager, or other team member outside of the operations team design scenarios for both nominal and off-nominal spacecraft behavior. Off-nominal scenarios should be based on issues seen during testing.

Every CubeSat has quirky behavior in one or more subsystems, and ORT scenarios should provide an opportunity to respond to previously observed subsystem misbehavior. For example, if the spacecraft radio has a tendency to freeze or reboot frequently, one ORT scenario should include a radio reboot during a ground station pass. If the attitude control system has known or suspected fault modes that could cause an uncontrolled tumble, the operations team should practice recognizing telemetry indicative of uncontrolled attitude and executing remedies to return the spacecraft to a safe, controlled attitude. ORTs should also include more common anomaly scenarios, like returning the spacecraft to nominal mode from safe mode. If time allows, ORTs should also include off-nominal behavior from the ground station as well as the spacecraft. The ground station team should be included in ORTs as well so that they can exercise procedures to detect faults in their equipment.

The ORT cycle should include several rounds of operations planning, simulated ground contacts with spacecraft telemetry and commanding, and after action debriefs and evaluation. After ORTs the team should assess any gaps in preparation or tools and work to address them before in-orbit operations begin.

2.2.3 *Timing of ORTs*

The delivery schedule and deployment schedule unique to each mission will determine the timing of ORTs. Missions that expect a long delay between delivery and flight could conduct ORTs during that schedule gap. If the CubeSat will be deployed from the International Space Station (ISS), there may be a delay of several months between launch and deployment, providing an excellent opportunity for team training and preparation. Launch and deployment dates are often uncertain however, so it is highly advisable to perform at least a minimal set of ORTs before the flight unit is delivered to the launch provider.

Budget realities often require a CubeSat team to stand down during wait times between delivery and launch or between launch and deployment. A refresher ORT 1–2 weeks before launch or deployment is advisable to dust off operations skills, tools, and knowledge of spacecraft idiosyncrasies. This prelaunch/predeployment ORT is also a final opportunity to shake out bugs in ground tools.

2.3 *Tools, procedures, and documentation*

Successful operations begin during I&T. To the extent possible the team responsible for functional and environmental testing should use the same tools to interface with the spacecraft that the operations team will use during flight. Developing these tools and interfaces early allows debugging before the mission-critical first contact and commissioning phases. Similarly, command sequence generation tools should be developed and used in I&T and then transferred to operations.

Automation of common tasks via scripts is valuable for reducing personnel effort and removing potential sources of human error. In the sprint to assemble and test a CubeSat, automation tasks are sometimes dropped or deprioritized in the interest of getting to the finish line. Including automation of ground tools for command and sequence generation, telemetry parsing and visualization and other routine tasks should be included in the software development schedule and timed to be ready at the beginning of the I&T campaign. Leveraging existing tools and platforms such as Git and GitHub, OpenMCT,^b COSMOS^c and AIT^d and adapting them to the needs of the mission can save considerable personnel effort on ground tool development.

Documentation of software, procedures, and spacecraft idiosyncrasies goes hand in hand with early development of ground tools. The I&T and development teams should collaborate on a “User Guide” for the spacecraft. This living document should include the following:

- basic information about each spacecraft subsystem and interfaces;
- operations block diagram (an example from the ASTERIA mission is shown in Fig. 1);
- command dictionary;
- listing of telemetry channels with typical values;

^b <https://nasa.github.io/openmct/documentation/>.

^c <https://cosmosrb.com/>.

^d <https://ait-core.readthedocs.io/en/master/>.

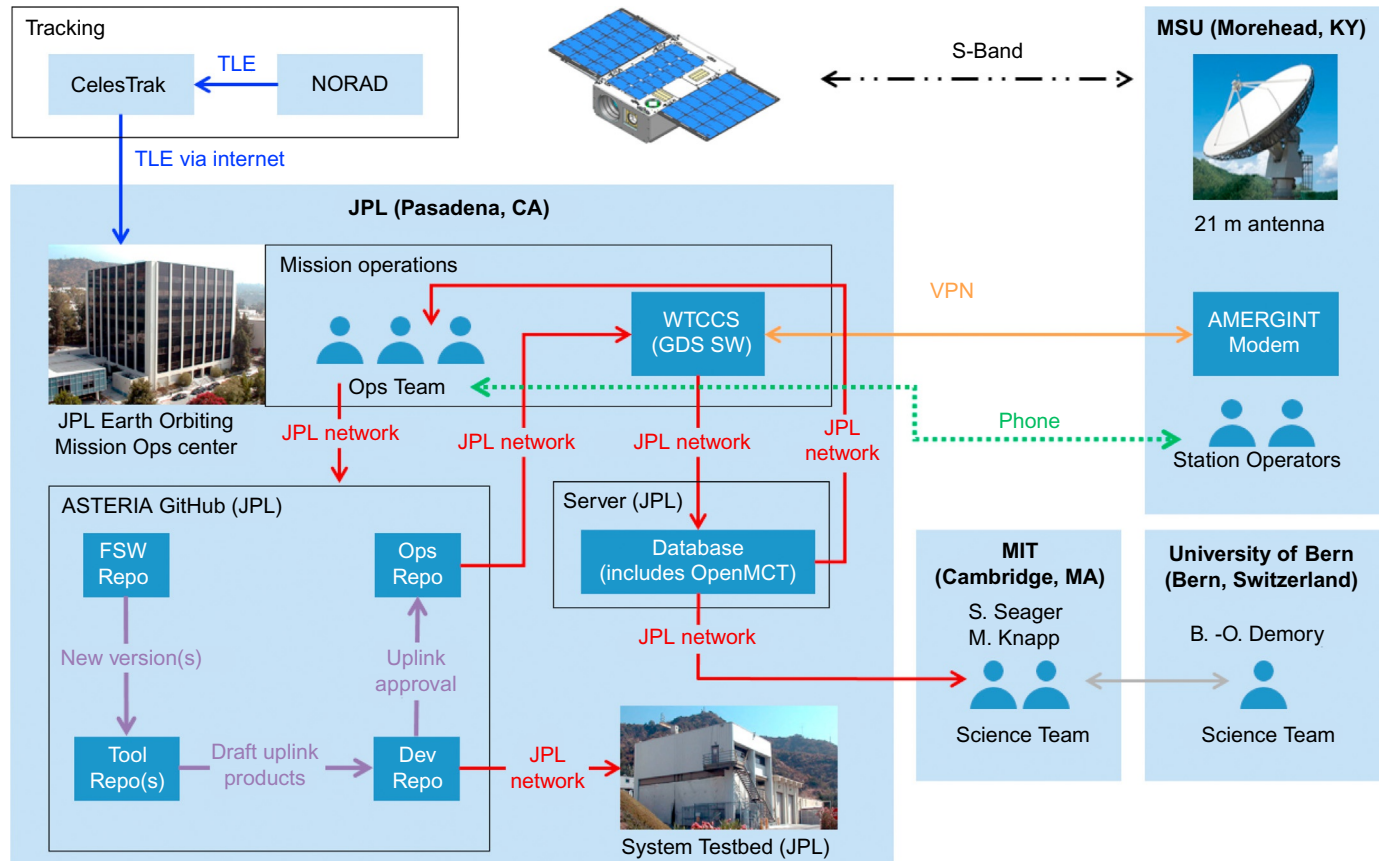


Fig. 1 Operations block diagram for ASTERIA [1–3]. The various modes of interaction between the ground station at Morehead State University in Kentucky, the operations center at JPL, and other mission stakeholders are shown by *solid and dotted lines*. Sequence generation and approval take place entirely in an institutional GitHub repository.

- Examples of both benign and serious flight software error messages;
- a thorough description of spacecraft and payload modes and a mode transition diagram;
- fault protection tables with descriptions of the responses to each fault;
- a listing of all watchdog timers, their duration, and the response if they trip;
- procedures for generating commands and/or sequences;
- data formats for telemetry and payload data;
- common anomalies seen in I&T;
- responses to correct common anomalies;
- listing and contact information for subsystem or component vendors.

The spacecraft User Guide should exist in a format that can be easily updated, such as wiki, and should be available to the I&T and operations teams. If operations are split between bus and payload, this User Guide should be available to both teams.

Like ground tools, documentation is sometimes, perhaps often, the first casualty of a budget or schedule crunch. One way to avoid this pitfall is to integrate documentation generation into the development process early so that it becomes a habit to update the User Guide regularly. The goal should be to grow documentation organically from many small contributions through mission development rather than to produce it in one chunk at the end of the development cycle.

3 First contact

Deployment is the moment of truth. The CubeSat that spent years in development and ground testing has been thrown in the deep end and must sink or swim on its own. It can be a difficult moment for both the development and operations team because there is nothing to be done but wait for a contact opportunity. The success of deployment and first contact rests primarily on preflight preparation and testing, including radio to ground station compatibility testing and fault protection design and testing.

Most CubeSat missions have a series of tasks that they must perform autonomously before ground contact. Many launch providers mandate a waiting period before the CubeSat turns on or releases any deployables. In some cases deployments happen as soon as the CubeSat exits the deployer. After deployment the spacecraft must turn on, charge batteries, and perform any deployments necessary for first contact, such as antennas or solar panels. The CubeSat should boot up into a minimal power mode that can be sustained for a long period of time to give the operations team ample opportunity to make contact and establish commandability.

First contact attempts are high-stress situations for the operations team, so thorough preparation is key. First contact scenarios should be practiced during ORTs so that the operator at the console for in-space first contact has developed a “muscle memory” of how to respond to nominal and off-nominal cases. The operations team should have a plan for the first week of contacts with the spacecraft that includes clearly defined decision gates for moving from one activity to the next. For example, there should be a checklist of telemetry values for commanding the spacecraft to nominal mode from its deployment safe mode. More than one fully trained operator should be present for these critical first contacts.

Personnel from the I&T, flight software, and fault protection teams should be present or on call for first contact and the commissioning phase that follows ([Section 4](#)). These are the people who know the spacecraft most intimately and who will be able to recognize anomalous states quickly. If at all possible, members of these teams should be part of the ops team for first contact and commissioning. The ground data system should be able to generate and distribute telemetry summaries and plots quickly so that the operations team and supporting personnel can use that information to assess the state of all spacecraft subsystems.

The stress of first contact can and should be mitigated with careful human factors design. Naturally the whole development team and other stakeholders may want to be present in the control room to experience the excitement of first contact. A crowd of people around the operations console adds considerable stress to the operator whose hands are on the controls. The environment around the operations point person should be kept calm and professional to keep nerves in check, so it is advisable to provide another viewing venue for the rest of the team that removes them from the immediate vicinity of the command console. An essential set of 2–4 subject matter experts should be present to support the operator in evaluating telemetry.

4 Commissioning

4.1 *Expect the unexpected: dealing with anomalies*

The commissioning phase of the mission is the period of time when each spacecraft subsystem, including the payload(s), is brought online and checked for functionality. It is during this period that the operations team should expect the unexpected, or in other words, anomalies. Each CubeSat is unique and therefore has its own unique anomalies, but we offer some general advice on detecting and responding to unexpected behavior in operations.

4.1.1 *Life signs: interpreting the received signal*

The first “life sign” of the spacecraft is the RF signal itself (or lack thereof). Thorough preflight testing ([Section 2.1](#)) should provide the operations team with intuition about how the flight radio behaves with the ground system. During operations, it is helpful to have a live frequency versus amplitude spectrum plot and/or time versus frequency spectrogram available to the operator during ground contacts. The data from these plots should be logged so that it can be viewed or played back at a later time for diagnostic purposes. A constellation plot showing IQ is also helpful for visually assessing the quality of the received signal from the spacecraft. These plots may be generated by either the ground station or the mission operations center (if they are not colocated) but should be available to the whole operations team.

Anomalies may first appear as a change in the strength or shape of the signal on the spectrum or spectrogram plot. The most extreme case is the lack of any detectable signal at all. The operations team should have a clear understanding of the fault scenarios in which the spacecraft is not transmitting when it is expected to be.

The operations team should practice walking through different fault scenarios and identifying other pieces of evidence that can help distinguish one scenario from another.

The following are example set of questions to work through when no signal is detected during a ground contact:

1. Is the ground station functioning properly? Can the ground station verify that it passes its own end-to-end functionality tests?
2. Is the Two Line Element (TLE) defining the orbit out of date or could the spacecraft be misidentified?
3. Could the lack of signal be a symptom of a recent flight computer reboot? How long does the flight computer take to reboot? How long does it take for the radio to be functional and ready to transmit after a flight computer reboot?
4. Does safe mode include a duty cycle for the radio to save power? How long is that duty cycle in each spacecraft mode? How many contact opportunities could be missed in a row in each mode given the duty cycle?
5. In what attitude scenario would the spacecraft transmit antenna be pointed away from the ground station?

Each mission is different, so the aforementioned questions are intended to be representative. Each operations team should know their spacecraft's unique fault tree pathways. If the spacecraft signal is seen by the ground station but its strength or shape is off nominal, there is a different fault tree for the operations team to traverse. In this case it is particularly important to preserve the data for other subject matter experts, like the communications team, to examine after the pass.

A backup ground station, or ground station network, is invaluable for disentangling ground station problems from spacecraft problems (refer to [Section 2.1.2](#)). If the spacecraft can be contacted by the backup ground station but not the primary, the problem clearly lies with the primary ground station. Even if a backup ground station is not included in the communication license, ghost tracks can be used for diagnostic purposes. A ghost track is when a secondary ground station tracks a spacecraft in receive-only mode while the primary ground station is attempting two-way communication. The ghost-tracking secondary station can determine whether the spacecraft is transmitting, whether the spacecraft signal power and spectral shape is nominal, and may be able to record and demodulate real-time telemetry packets. Just a few real-time telemetry packets can be vital for assessing spacecraft health.

4.1.2 Real-time telemetry: Reading the tea leaves

Real-time telemetry ([Section 2.1.3](#)) received during a ground contact is key to identifying spacecraft anomalies that do not affect the radio. Again the anomalies encountered will be mission specific, but in general, it is important for operators to have a clear understanding of nominal telemetry values so that off-nominal behavior can be identified quickly. The responses to off-nominal behavior will vary, but one frequently used response is to power cycle the offending subsystem or reset the entire spacecraft. The I&T team should thoroughly exercise power cycling of all subsystems and full system resets so that the behavior and timing is well understood. This testing will give the operations team more confidence in commanding subsystem power

cycles or spacecraft resets when or if they are needed. Quick decision-making may be required if, for example, the attitude control system is in a bad state and is not pointing the solar panels at the Sun. An in-pass reset could save the mission in such a scenario.

A flight-like testbed or FlatSat is essential to diagnosing anomalous telemetry. The testbed provides access to all telemetry, not just real-time telemetry, which aids in diagnosing issues. This is especially important if the telemetry received from the spacecraft is severely limited by communication problems. Sequences or commands that led to anomalous behavior can be run on the testbed to reproduce the observed behavior. Fixes can (and should) be run through the testbed before uplink to the spacecraft as well.

4.2 Human factors

4.2.1 Operation user interface

Quickly interpreting real-time telemetry during a ground contact is mission critical, so the way in which that information is displayed is critical as well. The operator's view of incoming telemetry and other data should include color coding of telemetry channels and visual alarms for off-nominal values. The user interface at the operator's station should be iterated during ORTs to minimize distraction and highlight the most important information.

During commissioning, at least two people should be present for ground contacts. The first operator is the point person and is responsible for sending commands. The second operator is a second set of eyes on the telemetry stream and can offer a second opinion if a quick decision is needed. Operator pairs should train together during ORTs to establish a good working relationship.

4.2.2 Healthy team, healthy spacecraft

After first contact the first 1–2 weeks of operations is most stressful for the operations team. Operations staffing should be frontloaded to the extent possible within budget constraints to spread out the work and allow for adequate rest. Many team members may be eager to put in extra hours or skip days off because they are excited that their CubeSat is finally flying, but project management should strongly encourage or require rest days and moderate working hours. Ensuring that operators have sufficient sleep and days off is essential to the safety of the spacecraft—tired operators make mistakes, and mistakes in anomaly situations can end a CubeSat mission prematurely. Underfunding and overworking the operations team are just as risky for the success of a mission as skipping essential preflight testing.

4.2.3 Human error

Human error is inevitable. The human factor considerations discussed in previous sections are aimed at minimizing operator mistakes, but some mistakes will happen anyway. Mitigation for human error has three components: (1) fault protection that

is robust to command or sequence errors, (2) thorough training and documentation, and (3) a solution-focused process for detecting and addressing errors when they do occur.

Fault protection design is complete and tested at the time of launch but should allow for updates as previously unknown fault modes come to light during operations. The second and third components of human error mitigation can be partially addressed by recording ground station contacts. Recording and archiving can be achieved through screen recording and/or video recording. Ground station pass recordings should be available to the operations team and wider support team as necessary. Pass recordings can serve as training tools for new operators; recordings where an anomaly was detected and handled are particularly valuable. Trainee operators can see what experienced operators see, but without the pressure of making split-second decisions.

Recordings of operator errors are valuable as a training tool as well but should not be used to point fingers or shame and blame an operator. Mistakes are an opportunity to uncover weak spots and vulnerabilities in procedures and tools that can then be remedied. The team should avoid questions like “Whose fault is this?” and “Why did you miss that?” and instead ask “What check can be built into our tools/procedure to prevent this from happening again?” and “What can we change in the user interface to avoid this problem in the future?” An increase in operator errors should be a red flag to the team that the operations staff may be overworked or burned out; corrective action should be taken immediately to lighten the load.

4.3 Mission assurance during operations

Members of a small CubeSat development/operations team often wear several hats. During operations, at least one team member should be designated the mission operations assurance manager (MOAM). On large missions the MOAMs are separated organizationally (in terms of their direct reports) from the main project organization so that they can be an independent voice without pressure from project management. A fully separate and independent MOAM is rarely practical on a small CubeSat team, but the role remains critical for maintaining a healthy spacecraft during operations. At minimum the MOAM and the project or mission manager should not be the same person.

The MOAM’s job is to provide the team with risk-informed decision-making guidance. Even if the MOAM is part of the development team, they should strive to provide an independent voice that focuses on maintaining spacecraft health. It is therefore important that the person wearing the MOAM “hat” is free to speak up and slow down or stop actions that he or she deems risky to spacecraft health and that the MOAM’s independence and authority are recognized and supported by project management. If the MOAM says “stop,” the team needs to listen.

The MOAM should take the lead in cross-checking commands and sequences, evaluating automating and other tooling, and overseeing operator training. Additionally, the MOAM should lead anomaly investigations and take responsibility for

tracking anomalies and their resolution. On the ASTERIA mission the MOAM was also in charge of fault protection design and mission assurance during the spacecraft development (see Ref. [4]).

5 Prime mission and beyond

5.1 Streamlining and optimizing operations for sustainability

As a mission progresses from the commissioning phase to nominal operations, the operations team will shift gears from “firefighting” to “routine maintenance.” In the commissioning phase the primary goal is to ensure that all subsystems, the payloads, and the overall spacecraft are healthy and functioning as intended. During the prime mission the operations team focuses on achieving the program’s key scientific and/or technical goals.

The operations team may shrink due to budgetary or other constraints after the commissioning phase of the mission. *At least one person should remain full time on operations during the prime mission and any mission extensions.* A full time operations lead provides continuity to the mission, training for new operators, and institutional knowledge of the spacecraft and its quirks. Though the operations budget may become increasingly squeezed as the program moves beyond the prime mission, a dedicated operations lead increases the chance that the spacecraft will continue to function and be available for further mission extensions and new experiments and observations if funding is available.

Automation is key to stretching a thin extended mission operations budget. Though ideally automated tools and procedures will already exist, further automating telemetry checks, commanding, and other labor-intensive tasks can reduce or eliminate the need for an operator to be physically present at the operations station during at least some ground station contacts. As operations become routine, longer blocks of sequenced commands can be generated so that the spacecraft can go longer without human-in-the-loop ground contacts.

5.2 Rebalancing risk posture

The risk posture of the mission and therefore operations may change if there is a follow-on extended mission after the prime mission ends. During the prime mission the risk posture is relatively conservative because the driving objective is to complete all of the mission goals before the end of the prime mission. Preserving the health of the spacecraft is paramount to that goal. After the prime mission objectives have been accomplished, it may be advisable to relax the risk posture to fully explore the capabilities of the spacecraft. For example, flight software updates are risky and potentially mission-ending so are often not attempted during the prime mission except in dire circumstances. In an extended mission however, a flight software update to fix non-critical bugs and/or add capabilities is a more reasonable risk to take. The MOAM

and project management should be deliberate about defining the extended mission risk posture so that they can provide clear guidance to the operations team.

5.3 Training new operators

CubeSat teams are known for high turnover, especially in university settings, so training new operators is necessary for mission longevity beyond the prime mission. Reviewing documentation is the first step in new operator training, so the mission “Users Guide” (Section 2.3) should be complete and up to date. Next, new operators should apprentice with experienced operators by sitting in on ground station contacts and observing. Pass recordings (Section 4.2.3) can also be used for this purpose. After becoming familiar with the workings of the spacecraft and pass logistics, the operations trainee can work through ORT-like simulated passes. If personnel resources for running simulated passes are limited (or nonexistent), playback of telemetry from both nominal and anomalous passes can be substituted. Finally the trainee should serve as the primary operator with an experienced operator observing for several ground contacts.

6 Looking to the future

The CubeSat commercial sector is rapidly evolving, as is the range of available operational models. Several CubeSat bus providers are offering ground station and operations support as part of their package, which is nudging the community toward a “split” operations model (Section 2.2.3). The increasing availability of commercial ground station networks is also changing the model for interaction between the mission team and ground station. While the specific implementations will change, thorough testing and documentation, as well as wide-open lines of communication between all parties, will remain the bedrock of successful CubeSat operations.

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